

Kai Barker

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ABOUT

Data Science senior at UC Santa Barbara with hands-on experience in statistical research, management consulting, and financial data analysis. Skilled in Python, R, SQL, and machine learning, with a track record of turning large complex datasets into clear decisions and measurable outcomes. Seeking a data analyst, data scientist, or analyst role where I can contribute immediately and grow.

EDUCATION

University of California, Santa Barbara

Statistics and Data Science B.S.

Santa Barbara, CA
Graduating June 2026

Major GPA: 3.90/4.0

Relevant Coursework: Machine Learning, Time Series, Stochastic Processes, Big Data, Advanced Statistical Models, Computational Science, Differential Equations, Technology Management, Probability and Statistics, Problem Solving in Computer Science

Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

EXPERIENCE

Engagement Manager

Consult Your Community

Santa Barbara, CA
February 2026 – Present

- Lead a team of 6 consultants on a 16-week EdTech engagement, defining 2 core project workstreams and serving as the primary liaison between client stakeholders and the analyst team.

Business Analyst

Consult Your Community

Santa Barbara, CA
October 2025 – Present

- Produced acquisition readiness assessment across operations, financials, and marketing for a small business client; identified 3+ areas to reduce owner dependency ahead of sale.
- Diagnosed over-reliance on owner relationships and gaps in client follow-up; recommended CRM and communication improvements projected to increase annual revenue by \$40,000 and improve business attractiveness to buyers.
- Built 10+ Tableau visualizations to highlight operational gaps, driving recommendations presented to client stakeholders.

Undergraduate Researcher

Department of Statistics and Applied Probability

Santa Barbara, CA
September 2025 – Present

- Analyze 70,000+ U.S. Senate stock trades in Python to test for statistically significant excess returns relative to market benchmarks.
- Calculate risk-adjusted performance using Sharpe ratios, finding members achieved 1.97 vs. market benchmark of 0.79
- Developed a data cleaning pipeline using FIFO matching and fuzzy matching to validate trade reporting accuracy, reconcile transaction records, and filter incomplete STOCK Act disclosures.

Research Assistant

Foundations of Financial Technology Lab

Santa Barbara, CA
September 2024 – Present

- Analyzed 20M+ Ethereum blocks and 100M+ wallets to quantify wealth distribution, calculated a Gini coefficient of 0.96.
- Produced visualizations tracking inequality evolution over 5 years, revealing the top 1% of wallets control 95%+ of Ether supply.
- Filtered institutional and Sybil addresses to isolate organic wallet behavior and improve the accuracy of distribution metrics.

PROJECTS

TrailForge AI Trail Recommendation Application

SB Hacks XII Hackathon

Santa Barbara, CA
January 2026 – January 2026

- Built a full-stack web app in 24 hours with OAuth 2.0, Python backend, and JavaScript/HTML frontend, integrating Strava, Google Search, and Gemini APIs for personalized trail recommendations.
- Engineered LLM prompts, designed a responsive UI with embedded trail links, deployed on Render within the hackathon window.

NeurIPS Ariel Data Challenge

Competition - ESA Ariel Mission

Santa Barbara, CA
July 2025 – September 2025

- Developed ensemble machine learning models (XGBoost + Gaussian Process Regression + Ridge) to predict exoplanet atmospheric compositions from 200+ GB of noisy spectral telescope data.
- Preprocessed 14,000 image files across 1,100+ exoplanets using signal processing and noise reduction techniques, achieving a GLL score of 0.311 across 283 wavelength channels.

MCMC & Simulated Annealing - Optimization Research

Academic Research

Santa Barbara, CA
May 2025 – June 2025

- Implemented and extended Metropolis-Hastings MCMC in Python to solve message decryption and the Traveling Salesman Problem, modeling both as permutation optimization over large discrete state spaces.
- Proposed a simulated annealing modification with an exponential cooling schedule, empirically validating a reduction in average decryption errors from 22.3 to 21.6 vs. baseline across 50,000 iterations.

SKILLS, CERTIFICATES & INTERESTS

Languages: Python, R, SQL, JavaScript, TypeScript, Java, C++, HTML, CSS

ML & Statistics: Scikit-learn, XGBoost, Gaussian Processes, SHAP, Feature Engineering, Statistical Modeling, Time Series

Data & Tools: Pandas, NumPy, Matplotlib, Tableau, Excel, Git, API Integration, LaTeX, Data Pipeline Development, PowerPoint

Certificates: JP Morgan Chase Software Engineering, JP Morgan Chase Quantitative Analysis, HackerRank SQL (ADVANCED)

Interests: Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball